

Gaze Pattern Genius: Gaze-Driven VR Interaction using Unsupervised Domain Adaption

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ABSTRACT

This research advocates shifting VR interaction to gaze-driven interaction, a more intuitive alternative to traditional controls like VR controllers or gestures. Our focus is on enhancing neural network recognition accuracy, especially with limited user-specific gaze data. We introduce a novel framework for capturing gaze gesture patterns and propose a template dataset concept to boost neural training. Our unsupervised domain adaptation model, blending template depth and sparse user data authenticity, consistently excels in recognizing gaze patterns across diverse users. Rigorous benchmarking against leading architectures consistently shows our method outperforming. Empirical user studies confirm: gaze-driven interactions not only elevate VR experiences but also redefine immersive VR control dynamics.

Index Terms: VR interaction—Gaze-driven interaction—Gaze gesture—Domain adaption;

1 INTRODUCTION

Enhancing VR experiences on Head-Mounted Displays (HMDs) heavily relies on intuitive human-computer interaction (HCI) [6]. Traditional methods [1, 4] like VR controllers, hand gestures, and voice commands suffice but can disrupt immersion or lack feasibility in certain scenarios. Developing more efficient, natural HCI approaches remains crucial.

While eye tracking excels in identifying focus areas in VR [3], intricate actions like turning virtual pages challenge its effectiveness, leading to a gap in exploring gaze tracking's broader applications beyond attention detection. The rapid speed of eye movements poses challenges in creating seamless interactions, hindering natural user reflection.

Collecting eye movement data is challenging, especially for new or specialized gestures, impeding reliable recognition models. A universal framework for eye movement data description across devices is needed to streamline data collection and focus on algorithmic improvements. Cross-domain learning challenges due to eye movement diversity further complicate progress.

To tackle these issues, we propose custom gaze-based patterns like alphabet input, employing a template dataset generation mechanism using control point transformation. We develop gaze-driven interactions in Virtual Reality Environments (VRE) through domain adaptation models, training networks efficiently with minimal gaze data to recognize gesture commands accurately.

Our main contributions can be summarized as follows:

- **Task.** Our work introduces a groundbreaking unsupervised domain adaptation approach that leverages GANs to facilitate

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gaze gesture interactions. The core objective is to address and overcome the inherent challenge associated with achieving elevated recognition accuracy, especially when neural networks are constrained by training on limited gaze data unique to individual users.

- **Dataset.** We introduce a versatile approach for describing gaze gesture patterns. Moreover, due to the scarcity of genuine user data requisite for training, we propose the creation of a template dataset to augment network training.
- **Method.** We propose a cross-domain learning model based on GAN. Through the fusion of a substantial template dataset with minute quantities of authentic user data, the model demonstrates an aptitude for recognizing gaze gesture patterns from diverse users. Comparative assessments against a range of sophisticated recognition networks reveal the superiority of our approach in terms of performance.

2 METHOD

We endeavor to propose a versatile framework to address the challenge of gaze gesture recognition for controlling VR interactions, wherein users are empowered to define their personalized gaze gesture patterns. We first collect data from predefined gaze gestures executed by the user wearing a VR HMD, forming what we define as the Target Domain. Then, we transform various template patterns into the Source Domain using two distinct techniques: the Straight Line Transform and the Bezier Curve Transform. Finally, we leverage both the Source Domain and the Target Domain to train a neural network, equipping it with the capability to effectively recognize gaze gestures.

Target Domain: Gaze Gesture Dataset. This section, we provide a gaze gesture pattern description method based on gaze ray data.

By employing an eye tracker, we can obtain the eye's position in three-dimensional space (x_0, y_0, z_0) and the gaze direction vector (i, j, k) at any given moment. During the recording of eye gestures within the time range $[T_a, T_b]$, we capture the gaze traces. If the data collection takes place in a three-dimensional VR environment, it becomes essential to establish the depth of the gaze plane (d) based on the type of scene. Conversely, when the collection is performed on a 2D screen, the screen itself acts as the gaze plane. We calculate the intersection points (x_t, y_t) between the gaze rays and the gaze plane at time $t \in [T_a, T_b]$.

We connect the obtained trajectory points in sequence on a virtual plane using line segments, resulting in a virtual plane trajectory diagram that serves as the representation of the gaze gesture pattern. We can determine the suitability of the current pattern for gaze gesture interaction based on the similarity between the obtained trajectory diagram and the original pattern diagram.

Source Domain: Template Dataset. Considering the high costs associated with eye-tracking data collection, we seek a concise method for data generation. This method allows users to craft

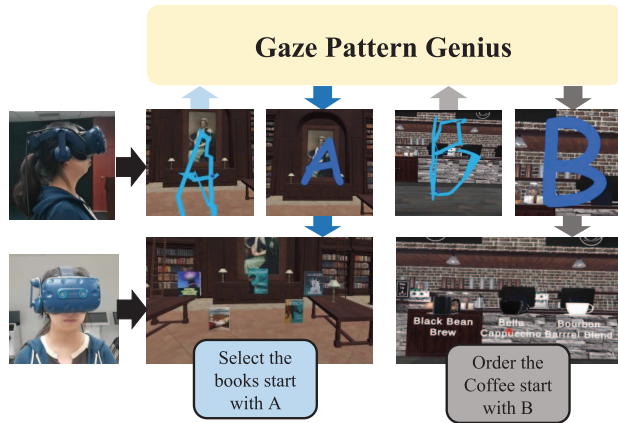


Figure 1: Our versatile framework allows users to complete gaze gesture patterns with a head-mounted display. The network model accurately identifies pattern categories and triggers interactions by providing feedback to the system.

their eye movement patterns effortlessly and generate corresponding dataset for training recognition networks. Thus, we present a template generation method that empowers users to custom-design their eye movement patterns, subsequently creating the associated dataset.

For trajectory patterns composed of straight line segments, we establish a set of positions for endpoints and inflection points, along with the sequence of line segment connections. Randomly perturb the endpoints and inflection points, and connect the new points according to the original connection sequence, and a large amount of new template data will be obtained.

We utilize Bezier curves [2] for template generation to automatically simulate the moving curves. We harness a Bezier curve visualization tool, enabling the precise manipulation of endpoints and control points. A complex pattern made of an assembly of Bezier curves with four control points results from this process. The positions of these pivotal points are systematically recorded and encapsulated within the point set. Randomly perturb the endpoints and control points, Generate a Bezier curve based on the new endpoints and control points to obtain new curve data.

Network Structure. Our Gaze Pattern Genius (GazePG) model is comprised of two distinct phases. The initial phase generates the source domain training dataset for the subsequent second phase. GazePG's training data mainly consists of two domains: the gaze pattern dataset and the template dataset. The gaze pattern dataset contains only 20 images for each class, while the template dataset consists of 4,000 images for each class used for training. The template dataset is to provide the foundation images for all the classes. We utilize the 26 letters of the alphabet as the 26 classes of gaze gesture patterns to be recognized. The recognition network is divided into four modules: the feature extraction module, the feature fusion module, the domain recognition module, and the category recognition module.

3 USER STUDY FOR COMPARISON WITH VR HANDLERS

3.1 Experimental Design

We compared gaze gesture and handle controller modes in a VR task scenario to assess their impact on immersion, cognitive load, and user preference. The objective was to determine if gaze gestures affect immersion and provoke simulator sickness symptoms compared to handle controllers. After a pre-study questionnaire, participants trained in both interaction techniques and completed 12 actions (twice each) using both methods. They rested 30 seconds between tasks and filled out six questionnaires (2 SSQs [5], 2 7-point

Likert scales and 2 IPQs [7]). Each subject conducted 48 trials, totaling around 60 minutes per experiment.

3.2 Participants

22 participants (9 males, 13 females; average age 20, SD = 1.78) were recruited. 8 had VR experience, 18 were myopic, and 7 wore glasses.

3.3 Discussions

Comparing gaze gestures to handle controllers, both maintained immersion and realism. Gaze gestures excelled in spatial immersion, fostering better spatial presence. They also enhanced user engagement and were preferred despite potential simulator sickness effects. Gaze gestures facilitated natural interaction target selection without significantly increasing cognitive load or inducing substantial sickness symptoms.

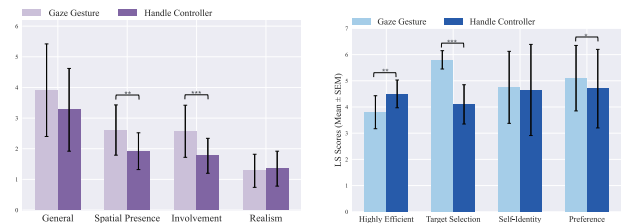


Figure 2: Subject scales. Left: IGroup Presence Questionnaire. IPQ scores are summarized across four domains compare. Error bars represent standard deviation. (*p < .05, **p < .01). Right: Scores of the LS subscales. Error bars represent standard deviation. (*p < .05, **p < .01, ***p < .001).

4 LIMITATIONS AND FUTURE WORK

The gaze gesture mode excels in immersive experiences and target selection, but hand-centric interactions favor the controller mode in specific scenarios. Blink-triggered interactions, while convenient, may cause eye discomfort with frequent movements, urging exploration of wider eye movement patterns.

Future research aims to detect and improve user misdrawings and enhance content distinctiveness. While our study covers 26 alphabet patterns, gaze gestures hold potential beyond these. Exploring additional patterns aligned with VR scenarios and user gaze patterns is a priority for upcoming research.

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